

JEE MAIN REVISION MASTER: COMPOUNDS CONTAINING NITROGEN

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Basicity of Amines & Gabriel Phthalimide Synthesis	14 PYQs	Amine Basics
High	Diazonium Salts & Carbylamine Test	13 PYQs	Diazonium & Tests

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Basicity of Amines & Gabriel Phthalimide Synthesis	$\text{Basicity in water: } (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{NH}_3 \quad (2^\circ > 1^\circ > 3^\circ > \text{NH}_3)$
Diazonium Salts & Carbylamine Test	$\text{Diazotization: } \text{Ar-NH}_2 + \text{NaNO}_2 + 2\text{HCl} \xrightarrow{0-5^\circ\text{C}} \text{Ar-N}_2^+ \text{Cl}^- + \text{NaCl} + 2\text{H}_2\text{O}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Basicity of Amines & Gabriel Phthalimide Synthesis (High Yield — 14 PYQs)

Core Principle: Interplay of inductive, steric, and solvation factors on amine basicity in gas vs aqueous phases, and Hoffmann bromamide degradation.

■ **Caution / JEE Pitfall:** In gas phase, basicity order is purely inductive: $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$. In aqueous solution, solvation and steric factors disrupt this order.

■ **Speed Trick:** Gabriel phthalimide synthesis yields PURE aliphatic primary amines (1°); aromatic amines cannot be prepared because aryl halides do not undergo $\text{S}_\text{N}2$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Basic strength of alkylamines in aqueous phase is not decided by _____." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "Which of the following reaction/s does not yield an amine?..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Biologically active adrenaline and ephedrine used to increase blood pressure contain..." [Ans: N/A]

3.2 Diazonium Salts & Carbylamine Test (High Yield — 13 PYQs)

Core Principle: Preparation of benzene diazonium chloride (0-5°C), Sandmeyer/Gattermann halogenation, azo coupling dyes, and Hinsberg separation.

■ **Caution / JEE Pitfall:** Carbylamine test gives foul-smelling isocyanide ONLY with primary amines (1°); secondary and tertiary amines give NO reaction.

■ **Speed Trick:** Hinsberg reagent (C₆H₅SO₂Cl): 1° amine gives product soluble in KOH; 2° amine gives product insoluble in KOH; 3° amine does not react.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Organic compound 'D' is..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Nitration of aniline in strong acidic medium gives significant amount of m-nitroaniline because..." [Ans: B]
- [2025 Shift] Q#2 (MCQ): "Match the compounds given in List - I with the items given in List - II. List - I List - II (I) Benzenesulphonyl Chloride (a) Zwitterion..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Organic compound 'D' is

- (A)
- (B) CH₃COOH
- (C) CH₃CHO
- (D)

Correct Answer: None

Solution Analysis: Benzene first reacts with concentrated HNO₃ and concentrated H₂SO₄ at 323-333K. This is nitration of benzene, and the product formed is nitrobenzene, marked as (A). C₆H₆ + HNO₃ / H₂SO₄ → C₆H₅NO₂ (A). Nitrobenzene (A) is then reduced using Fe/HCl. The nitro group, -NO₂, changes into an amino group, -NH₂. So the product is aniline, marked as (B).

$$\text{C}_6\text{H}_6 + \text{HNO}_3 \xrightarrow{\text{H}_2\text{SO}_4} \text{C}_6\text{H}_5\text{NO}_2 \xrightarrow{\text{Fe/HCl}} \text{C}_6\text{H}_5\text{NH}_2$$

Benchmark Problem 2: KCET 2026 (Q#2)

Basic strength of alkylamines in aqueous phase is not decided by _____.

- (A) Inductive effect
- (B) Solvation effect
- (C) Steric hindrance
- (D) Hyperconjugation effect

Correct Answer: None

Solution Analysis: Basic strength of alkylamines in aqueous phase is not decided by hyperconjugation effect. So, the correct option is: Option D: Hyperconjugation effect. In aqueous solution, the basic strength of alkylamines mainly depends on: Inductive effect: Alkyl groups show +I effect and increase electron density on nitrogen. Solvation effect: The conjugate acid formed after accepting H⁺ is stabilised by water molecules. Steric hindrance: Bulky alkyl groups can reduce proper solvation and affect basicity. But hyperconjugation is not an important factor for deciding the basic strength of alkylamines.